Fiscal Year:	FY 2023	**Task Last Updated:**	FY 05/10/2023
PI Name:	Smith, Scott M Ph.D.		
Project Title:	Space Biochemistry Profile		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Bone Fracture**:Risk of Bone Fracture due to Spaceflight-induced Changes to Bone (2) **Food and Nutrition**:Risk of Performance Decrement and Crew Illness Due to Inadequate Food and Nutrition (3) **Immune**:Risk of In Mission Impacts, Adverse Health Events or Long-Term Health Impacts due to Altered Immune Response (4) **Nutrition**:Risk of Inadequate Nutrition (5) **Osteo**:Risk Of Early Onset Osteoporosis Due To Spaceflight (6) **Renal Stone**:Risk of Renal Stone Formation		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	scott.m.smith@nasa.gov	**Fax:**	FY 281-483-2888
PI Organization Type:	NASA CENTER	**Phone:**	281-483-7204
Organization Name:	NASA Johnson Space Center		
PI Address 1:	Biomedical Research and Environmental Sciences Division/SK3		
PI Address 2:	2101 NASA Pkwy		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058-3607	**Congressional District:**	36
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2012 Crew Health NNJ12ZSA002N
Start Date:	08/01/2013	**End Date:**	09/30/2022
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 9/30/2022 per HRP HHC element and PI (Ed., 7/8/21) NOTE: End date changed to 12/31/2021 per PI (Ed., 2/25/21) NOTE: End date changed to 12/31/2020 per PI (Ed., 5/10/19) NOTE: End date changed to 12/31/2019 per PI (Ed., 6/5/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Zwart, Sara Ph.D. (University of Texas Medical Branch/) Heer, Martina Ph.D. (University of Bonn, Germany)		

Grant/Contract No.:	Internal Project
Performance Goal No.:	
Performance Goal Text:	
Task Description:	As long-duration spaceflights continue and the operational suite of countermeasures is modified, the food system is updated, and the duration of missions lengthens, it will be important to evaluate and monitor a broad set of biomarkers for key physiological systems. The Nutritional Status Assessment Supplemental Medical Objective (aka “Nutrition SMO”) was initiated in 2006, and has yielded significant clinical, operational, and research data. This proposal aims to extend the Nutrition SMO, under the guidelines provided in the NRA (NASA Research Announcement). Nutrition SMO data have been used to help identify or explain medical, scientific, and even engineering issues that have occurred during or after International Space Station (ISS) missions. The data have been used by Medical Operations on multiple occasions, to confirm the effectiveness of vitamin D supplementation, to test for nutrient toxicities (secondary to supplement use), to evaluate blood and urine chemistries after instances of kidney stones and gout symptoms in crewmembers, and to evaluate the effects of using a new exercise device on bone and calcium metabolism. The ISS Program Office has used these data to determine factors contributing to the Urine Processor Assembly failure and to make forward operational decisions. Perhaps most striking, the data provided evidence that one-carbon metabolism may be altered in crewmembers who experienced vision changes post flight, the highest Human Research Program risk. The relationship between nutritional status and 1-carbon metabolism would likely never have been discovered if the Nutrition SMO were not being conducted. The impact of the data collected to date provides a strong rationale for continuing with an updated version of this protocol, eliminating some tests while expanding others, to provide a repository of data to other scientific Disciplines. We have extensive experience with these types of analyses, sample and data management, transfer to data archives, and data reduction for medical, management, and research purposes.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The findings from this study will help us better understand physiological adaptation to spaceflight, and will help evaluate countermeasure effectiveness. These results help the ISS and Artemis Programs, the Human Research Program (HRP), Space Medicine, Office of the Chief Health and Medical Officer (OCHMO), NASA engineers, individual astronauts, and other experiments. These results will also inform the general, medical, and scientific communities on human health and physiological issues in an altered gravity environment. There have been and dontinue to be significant implications of these findings.
Task Progress:	This protocol was terminated after Expeditions 56/57 (E56/57), ending collection of valuable data with impact to the International Space Station (ISS) Program, NASA Human Research Program (HRP), Office of the Chief Health and Medical Officer (OCHMO), Space Medicine, the Environmental Control and Life Support System (ECLSS), other investigations, and to individual astronauts. Extended sample analyses continue, along with reviewing, compiling, and presenting/publishing data.
Bibliography Type:	Description: (Last Updated: 01/19/2026)
Articles in Peer-reviewed Journals	Gabel L, Liphardt A-M, Hulme PA, Heer M, Zwart SR, Sibonga JD, Smith SM, Boyd SK. "Incomplete recovery of bone strength, density, and trabecular microarchitecture at the distal tibia 1 year after return from long-duration spaceflight." Sci Rep. 2022 Jun 30;12:9446. https://doi.org/10.1038/s41598-022-13461-1 ; PMID: 35773442; PMCID: PMC9247070 , Jun-2022
Articles in Peer-reviewed Journals	Stroud JE, Gale MS, Zwart SR, Heer M, Smith SM, Montina T, Metz GAS. "Longitudinal metabolomic profiles reveal sex-specific adjustments to long-duration spaceflight and return to Earth." Cell Mol Life Sci. 2022 Nov 1;79(11):578. https://doi.org/10.1007/s00018-022-04566-x ; PMID: 36319708 , Nov-2022
Articles in Peer-reviewed Journals	Lee SMC, Ribeiro LC, Martin DS, Laurie SS, Zwart SR, Chen X, Young M, Crucian B, Smith SM, Macias BR. "Arterial structure and function in the years after long-duration spaceflight." J Appl Physiol (1985). 2025 Jun 1;138(6):1474-88. https://doi.org/10.1152/japplphysiol.00264.2024 ; PMID: 40304245 , Jun-2025
Articles in Peer-reviewed Journals	Matheson BE, Walle M, Liphardt A-M, Hulme PA, Heer M, Zwart SR, Sibonga JD, Smith SM, Gabel L, Boyd SK. "Recovery of bone microarchitecture and density four years after spaceflight: Two case studies." npj Microgravity. 2025 Jul 28;11(1):47. https://doi.org/10.1038/s41526-025-00511-x ; PMID: 40721429; PMCID: PMC12304210 , Jul-2025